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Remedial Engineering Report

Date 15.07.2026
To The Owners of Strata Plan Number 17078
Attention Mr Thomas Louw

**Remedial Engineering Report #1 at
17a Fort Street, Petersham NSW 2049
Strata Plan Number 17078**

Dear Owners,

1.0 Introduction

Further to the request of Thomas Louw on behalf of the Owners' Corporation of Strata Plan 17078, Mr Rinay Singh and Mr Spencer Llewellyn, Remedial Team Leader and Remedial Engineer respectively, of Partridge Remedial Pty Ltd, consulting structural and forensic engineers, carried out an inspection on the 12th of June 2026. At the time of our inspection the weather conditions were fine.

The purpose of the inspection and this report was limited to conduct a visual inspection of the façade, rooftop waterproofing, staircase and water ingress into two units at the above property. Items to inspect include:

- (i) Water ingress into two units due to:
 - a. waterproofing,
 - b. roof detailing,
 - c. windows
 - d. cavity flashing
- (ii) Roof Waterproofing (including drainage)
- (iii) Façade elements
 - a. Windows
 - b. Façade Painting
 - c. Removal of redundant services
- (iv) Fire Stairs

This report lists our observations made during the inspection followed by our recommendations for the remedial works deemed necessary.

At the time of our inspection, access was not provided to Unit 4.



2.0 Description

The property is situated on the western side and northern side of Fort Street. For the purposes of this report the front of the building facing Fort Street is deemed to be facing North.

The property comprises of four (4) lots in total and is two (2) tall. The building is constructed of load bearing masonry walls which appear to support a timber framed flat roof. The roof is surrounded by rendered brick masonry balustrades a twisted steel spindle segment. Additionally, there are multiple tile clad patched pent roofs around the perimeter of the main flat roof. The façade is rendered and painted and features timber windows.

Please refer to Figure 1 below which illustrates an aerial view of the property, showing its immediate surrounds and approximate property boundaries shown in Red.



Figure 1 – Aerial Overview of the property (Image Courtesy of 'NearMaps')



3.0 Observations

3.1 Common Entry Stairs

- The entry to the units is via a common entry staircase with terrazzo treads and steel balustrade (refer to photo 1).
- Evidence of internal water penetration in the form of localised water staining and flaked paint to the ceiling, walls and cornice of the common staircase area (refer to photos 2, 3 and 4).



Photo 1 General view of common entry stairs



Photo 2 Water staining to the ceiling



Photo 3 Water staining to the cornice



Photo 4 Flaking paint and water staining to upper common stair walls



3.2 Unit 3

- Unit 3 appeared to be in largely good condition with the internal finishes being recently painted.
- Cracking was observed to the kitchen window glazing (refer to photo 5)
- Deterioration of the windows was observed externally with flaking/missing paint, corroded fixings and separation of timber frame elements (refer to photos 6, 7 and 8).



Photo 5 Crack to kitchen window glazing



Photo 6 Deteriorated to sunroom windows



Photo 7 Deteriorated to sunroom windows



Photo 8 Deteriorated to sunroom windows



3.3 Fire Stairs

- The fire stairs to the rear of the property are of concrete construction and feature two balustrades being what appears to be original steel balustrading constructed of steel pipe infilled with steel wire weave and newer powder-coated balustrades.
- Multiple instances of concrete spalling were observed to the soffit of the staircase.
- Evidence of further concrete spalling to the level 1 common door in the form of cracking to concrete lintel.
- The steel pipe balustrading features heavily corroding to all elements including the steel wire, steel piping and the fixings.
- There are multiple instances of the steel pipe adjacent the façade with corrosion pitting. This is likely the result of non-compatible original fixings accelerating corrosion as the pitting is adjacent what appears to be newer fixings.



Photo 9 General view of the fire stair

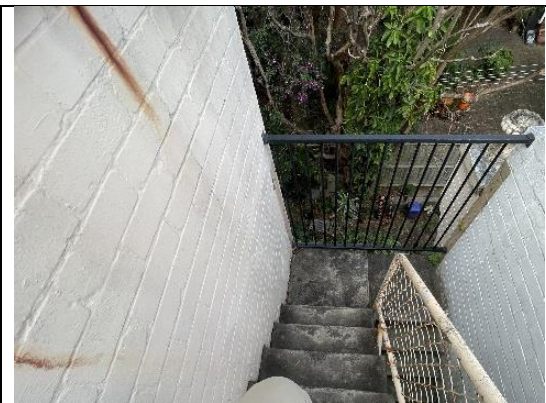


Photo 10 General view of the fire stair



Photo 11 Concrete spalling to staircase soffit



Photo 12 Concrete spalling to staircase soffit



Photo 13 Cracking to level 1 concrete lintel

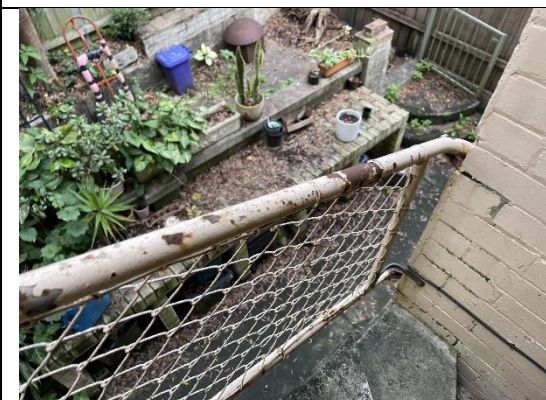


Photo 14 Corrosion to steel pipe balustrade toprail



Photo 15 Corrosion to multiple steel balustrade elements



Photo 16 Corrosion pitting to the end of the steel balustrade bottom rail



3.4 Roof

- Roof is a flat roof with a timber pitched roof pergola to the north eastern side and multiple timber framed pent roofs clad with tiles.
- The existing roof membrane appears to be overcoated with liquid membrane and laid upon a timber framed substrate. There are seams running the length and width of the roof which may be from being laid upon panels of fibre cement (fc) sheeting or similar, or from being overcoated on an existing sheet membrane.
- The roof features brick masonry balustrades a twisted steel spindle segment (refer to photo 17).
- The existing drainage is not acting in service with ponding to multiple locations and evidence of ponding to various other locations (refer to photos 18, 19 and 20).
- An instance of cracking to the edge of the membrane (refer to photo 21).
- Corroded fixings were observed to the brick parapet (refer to photo 22).
- Leaf litter collected within the drainage outlet (refer to photo 23).
- Patch repair to the pent roof above Unit 3 sunroom (refer to photo 24).



Photo 17 General view of roof with twisted steel spindle section.



Photo 18 Ponding water above common staircase



Photo 19 Ponding water above common staircase



Photo 20 Evidence of water ponding



Photo 21 Deteriorated waterproof membrane with cracking



Photo 22 Brick parapet with corroded fixings



Photo 23 Leaf litter filling outlet



Photo 24 Patch repairs to the pent roof above Unit 3 sunroom



3.5 Façade

- The building façade is largely rendered and painted and features timber windows and multiple small ledges and planterboxes.
- There is an instance of missing render behind the northern-most downpipe of the western façade (refer to photo 26).
- The façade features what appears to be several redundant services (particularly to the eastern façade) which are pitted and not visibly routed to any location (refer to photo 27).
- The windows to the façade are largely in a deteriorated condition with various instances of flaking paint, rot and corrosion (refer to photos 28, 29, 30 and 31).
- There is step cracking to the head of the door to the common laundry (refer to photo 32).



Photo 25 General view of render and painted façade



Photo 26 Missing render to western façade



Photo 27 Redundant services to the eastern façade



Photo 28 Flaking paint and corrosion staining to southern window



Photo 29 Rot and flaking paint to southern window



Photo 30 Rot to southern window



Photo 31 Flaking paint to Unit 4 sunroom window



Photo 32 Step cracking to common laundry lintel



4.0 Discussion and Recommendations

4.1 Fire Stair Condition

The fire stairs to the rear of the property are of concrete construction and feature two balustrades being what appears to be original steel balustrading constructed of steel pipe infilled with steel wire weave and newer powder-coated balustrades.

Concrete spalling was observed to the soffit of the staircase and evidence of concrete spalling was observed to the concrete door lintel in the form of cracking.

Concrete spalling is the deterioration of reinforced concrete elements, where sections of the concrete surface crack, delaminate, and ultimately detach from the structure. This process typically occurs as a result of corrosion of the embedded steel reinforcement, which is most commonly caused by prolonged exposure to moisture and oxygen penetrating through the concrete cover. As steel reinforcement corrodes, it expands in volume. This expansion leads to cracking, delamination and spalling of the concrete. Please see below diagram from HB84:2018 Guide to Concrete Repair and Protection (HB84) for a diagram of this occurrence.

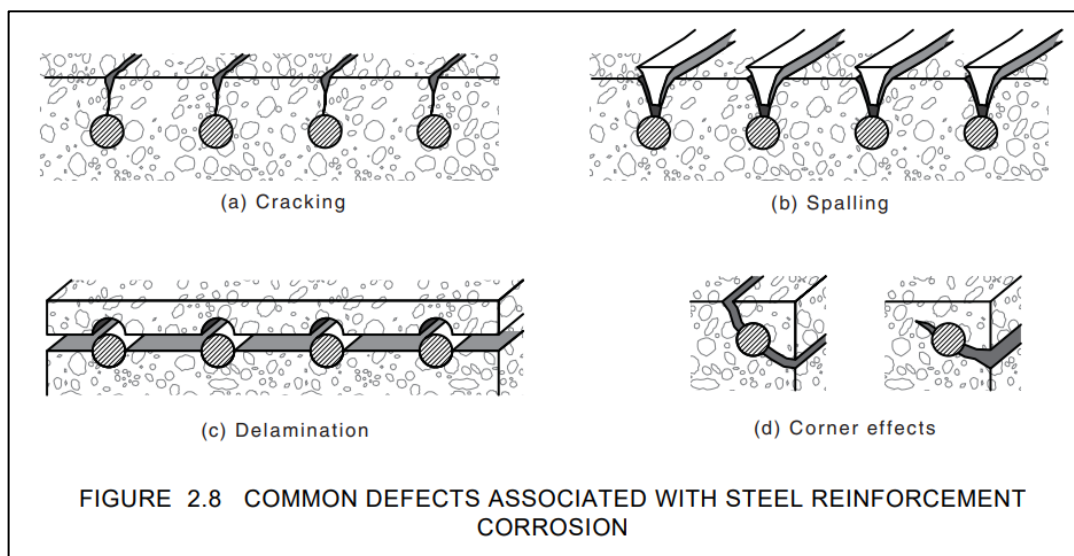


Figure 4.1: Extract from HB84 detailing external indicators of concrete spalling

Additionally, corrosion was observed to the steel pipe balustrade which has pitted the steel pipe completely through. It appears that attempted repairs have been undertaken in the form of additional fixings to attempt to maintain the balustrade integrity.

With continuous exposure to moisture and the environment, the steel balustrade, fixings, steel reinforcement and concrete will continue to deteriorate, reducing the structural integrity of these elements. This poses a significant risk to the building's stability and safety of its occupants in the long term if not addressed promptly. Further, remediation costs will increase over time.

Recommendation:

Typically, we would recommend concrete spalling repairs and the replacement of the balustrade. Further, the concrete should be protected with a suitable waterproof membrane.

We understand that the owners have considered replacement of the staircase with a new concrete or a prefabricated steel staircase. We expect that the construction of a new staircase would present more value for money in the long term.

We wish to emphasise that under the *National Construction Codes* (NCC), there are restrictions on how the risers and goings within the stairs can be constructed. Based on a preliminary review of the existing design, it is likely that a redesign of the would result in a change to the layout of the steps. The current goings of the stairs as well, as the slope relationship, are below the minimum. Whilst retrospectively altering the stairs to meet the NCC requirements is not necessary; it will be when newly constructed. This may lead to a spatial problem as the existing staircase is very heavily constrained by the existing stairwell and doors.

Based on the above, we also recommend the owners engage a BCA consultant to provide advise on the staircase fire compliance and the dimensions of the proposed new staircases goings and riser.



4.2 Roof waterproofing

The existing roof is a combination of a timber framed flat roof with multiple timber framed tiled pent roofs. The flat roof forms the majority of the rooftop and relies upon a channel drain along the western side of the roof to provide drainage. This is then serviced by rainheads on the western façade.

All roofs are required to feature falls that adequately drain water. There are multiple instances of ponding water and therefore the existing drainage to the flat roof area is not acting in service. This ponding presents a slip hazard to the common roof area. Further, waterproof membranes are not typically designed to remain submerged. The observed evidence of water ingress to the common stairwell is likely the result of ponding water pathing through a failure in the rooftop membrane or façade coating.

We note that whilst water ingress was not observed to Unit 3, and access was not granted to Unit 4, the lack of suitable falls to the roof will result in significant water ingress to the units below should the membrane fail.

During our inspection, we observed cracking to the nosing of the rooftop membrane at the entrance of the rooftop. This would result in water pathing through the substrate and into the building.



*Left: Cracking to the roof waterproof membrane.
Right: Water ponding to the roof.*

We also note that unreinforced masonry balustrades do not meet the current lateral loading requirements as per *AS1170.1:2002 'Structural Design Actions Part 1: Permanent, imposed and other actions'*. Further, the observed northern parapet balustrades twisted steel spindles do not meet the minimum requirements of the *NCC*, which stipulates that any horizontal element within the barrier between 150 mm and 760 mm above the finished floor level must not facilitate climbing, for a floor more than 4000mm above the surface beneath.

Recommendation:

We recommend the owners consider replacement of the roof waterproof membrane. Whilst no water ingress was observed to habitable areas, it is clear that the membrane is nearing the service life with failing details and a lack of falls. As part of these works, we recommend removal of redundant services, replacement of corroded fixings and correction of falls.

We note that the changes to building legislation from the Design and Building Practitioners Act prevent localisation repairs being a consideration. Accordingly, where rectification of falls and waterproof membrane repair is required, the full system is required to be remediated. Further, the new waterproofing system will need to be designed by a registered design engineer with a comprehensive scope of works and DBPA designs to ensure compliance with current codes and standards.

By undertaking this comprehensive approach, the underlying issues with the existing waterproofing membrane and drainage pipe issues can be addressed, ensuring long-term durability and compliance with the current standards. We recommend any consequential damage from water ingress to be addressed once waterproofing works have been undertaken.

Additionally, whilst retrospective replacement of the balustrade is not necessary, if any remedial works affect the integrity of the masonry balustrade, provision will need to be made to ensure that the balustrades are compliant with *AS1170.1* and the *NCC*.



4.3 Window Condition

The windows to the property are timber framed. Timber framed windows are unlikely to meet modern standards, as wind and water resistance requirements have become more stringent.

Further, timber windows rely on paint coatings to prevent degradation. Where the paint coating has not been maintained, dilapidation such as the observed flaking paint, rot and fixing corrosion is accelerated. Regular maintenance of windows should be included within the buildings' Capital Works Plan as maintenance of the windows prevents costly replacement.

Additionally, given the observed cracking to the Unit 3 kitchen window and the windows age, the windows are likely not tempered and therefore not up to modern safety requirements. Whilst retrospective replacement is not required, where the glazing has cracked the owners should consider replacement with a modern system including modern safety glass.

Recommendation:

Accordingly, we recommend replacement of any existing timber framed windows which have cracked glazing or rotted timber elements. We would recommend they be replaced with modern powder-coated aluminium windows which meet the current NCC and Australian Standards. We recommend these windows be installed with glazing compliant with modern standards which will provide superior safety, thermal and acoustic properties.

The owners may also consider replacement of all windows with modern powder-coated aluminium. Given the condition of the paint coating, the timber construction and the glazing, this would upgrade the building to be safer, easier to maintain, and superior thermally and acoustically.

4.4 Façade Condition

The façade appeared to be in generally sound condition besides the following:

- Missing render to western façade (refer to photo 26)
- Step-cracking to the head of the common laundry door. Given the location of the step-cracking we expect this is the result of subsidence. We recommend further monitoring for progression of cracking (refer to photo 32)
- Rusted fixings to services (refer to photo 22)
- Hole in southern façade (refer to photo 16)

We note that evidence of water ingress was observed into the common staircase in the form of water staining and flaking paint. This may be the result of water pathing through corroded fixings or cracks to the façade. Given the location, this may be from either the façade or the roof above. We provide further comment in Section 4.2 above.

Further, given the location of the missing render observed to the western façade, it is likely contributing to water ingress into Unit 4. We note that access was not granted to Unit 4 at the time of the inspection.

Recommendation:

We recommend repairs be undertaken including the following:

- Miscellaneous render repairs including to the western façade and to where fixings have deteriorated.
- Replacement of all corroded fixings (noting some may be fixed to redundant services)

Additionally, we recommend conducting water testing of the roof and the façade to confirm the source of water ingress to the common internal stairwell.

We note that whilst onsite we observed multiple services to the façade appear redundant. Various pipes were rusted through or were not routed into the building. Should the owners wish to remove any redundant pipework, a plumber will be able to test and confirm serviceability of all plumbing and remove redundant pipework.



5.0 Conclusion

We have inspected the following concerns at the request of Thomas Louw on behalf of the Owners' Corporation of Strata Plan 17078.

- (i) Water ingress
- (ii) Roof Waterproofing (including drainage)
- (iii) Façade elements
- (iv) Fire Stairs

Our report is limited to these areas. this is not considered a Specification and therefore cannot be used to tender out to a building contractor. Partridge can provide a Scope of Works or Specification for the recommended remedial works.

In summary, we recommend that Items within Section 4 are considered for rectification.

Should you or the Owners' Corporation have any further queries please do not hesitate to contact the undersigned

Yours faithfully,

Prepared by:

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Engineer, Partridge Remedial
For and behalf of Partridge Remedial Pty Ltd

Reviewed by:

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Remedial Team Leader, Partridge Remedial